The Havlík's Law Asymmetry: Evidence That Liquid-Context Jers Were Not Real Vowels

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Abstract

Havlík's law governs the fall of the jers in Slavic: strong-position reduced vowels (ъ, ь) vocalize to full vowels, while weak-position jers delete. This paper documents a categorical asymmetry in Czech outcomes that has not been explicitly identified in the literature: strong-position jers in non-liquid environments vocalize to /e/ in 100% of attested cases ($n = 18$), while strong-position jers adjacent to liquids (r, l) vocalize to /e/ in 0% of cases ($n = 27$). Instead, liquid-adjacent jers yield syllabic consonants (*vlk*, *smrt*, *prst*). The seven Modern Czech words that show /e/ near liquids (*červ*, *černý*, *blecha*, *pleť*) are demonstrated to be secondary epenthesis from the 13th–15th centuries, not Havlík's law vocalization (10th–12th c.): Old Czech had *črv*, *črný*, *blcha*, *plt*. Furthermore, when hard syllabic /l/ resolved in Czech, it produced /u/ (*žlutý*, *dlouhý*), not /e/—confirming a process distinct from jer vocalization. Cross-Slavic data from Serbian, Russian, and Polish corroborate the pattern. In Russian, where all liquid-context jers vocalized, the vowel quality reveals that front jer ь before /l/ yields /o/ rather than the expected /e/—matching the pattern of East Slavic polnoglasie (pleophony) rather than Havlík's law. We argue that liquid-context jers in OCS orthography were notational conventions for already-syllabic consonants, not real reduced vowels, and that Havlík's law applies only to genuine reduced vowels inherited from PIE short **i*/**u*. This reanalysis eliminates the need for an ad hoc liquid-context exception in Havlík's law and simplifies the diachronic account of Slavic syllabic consonants.

1 Introduction

The fall of the jers (падение еров) is among the most consequential sound changes in Slavic historical phonology. Proto-Slavic had two reduced vowels, the back jer ъ and the front jer ь, which by the 10th–12th centuries were either strengthened to full vowels or lost entirely, depending on their position in the word. The governing regularity is Havlík's law: counting from the end of the word, jers in odd-numbered positions ("weak") are deleted, while jers in even-numbered positions ("strong") vocalize to full vowels—typically /e/ in Czech and Slovak, /o/ or /e/ in Russian (see Havlík 1889; Shevelov 1964; Bethin 1998).

The law is remarkably regular for jers in non-liquid environments. Proto-Slavic *$s^{\text{ъ}}n^{\text{ъ}}$ 'sleep' yields Czech *sen*, *$d^{\text{ь}}n^{\text{ь}}$ 'day' yields *den*, *$p^{\text{ь}}s^{\text{ъ}}$ 'dog' yields *pes*—in each case the strong jer vocalizes to /e/ while the weak jer is deleted.

But for jers adjacent to liquids (/r/, /l/), the outcome is categorically different. Proto-Slavic *$v^{\text{ь}}lk^{\text{ъ}}$ 'wolf' yields Czech *vlk*—not **velk*. *$s^{\text{ъ}}m^{\text{ь}}rt^{\text{ь}}$ 'death' yields *smrt*—not **smert*. *$p^{\text{ь}}rst^{\text{ъ}}$ 'finger' yields *prst*—not **perst*. The strong-position jer does not vocalize to /e/; instead, it disappears, and the liquid becomes a syllabic consonant.

This asymmetry is well known descriptively: any Czech grammar notes that the language has syllabic /r/ and /l/. What has not been explicitly documented, to the best of our knowledge, is the **categorical nature** of the split and its implications for the phonological status of liquid-context jers. In this paper we present systematic data showing that the asymmetry is **total**—100% vocalization for non-liquid jers versus 0% for liquid-adjacent jers—and argue that this categorical split constitutes evidence for a fundamental distinction between two types of graphemes written identically as ъ/ь in OCS:

1. **Real reduced vowels** (from PIE short **u*, **i*), which participate in Havlík's law normally;

2. **Notational conventions for syllabic consonants** (from PIE syllabic resonants **l̥*, **r̥*), which were never real vowels and therefore have nothing to vocalize.

This reanalysis eliminates the need for an ad hoc exception in Havlík's law for liquid contexts and simplifies the diachronic account of Slavic syllabic consonants.

2 Background

2.1 Havlík's law

Havlík's law (Havlík 1889) is typically formulated as follows. In a sequence of jers counted from the right edge of the word, the rightmost jer is weak (position 1), the next is strong (position 2), and so on, alternating. A full vowel resets the count. Weak jers are deleted; strong jers are vocalized to full vowels.[1]

The law operates with near-exceptionless regularity across Slavic for jers in non-liquid environments (see §3.1 below). Its status as a genuine phonological regularity rather than a mere descriptive generalization is supported by modern experimental evidence: Gouskova & Becker (2013) showed that Russian speakers productively apply yer-like alternations to nonce words, with special constraints near sonorants.

[1] The specific reflex varies by branch: /e/ in Czech and Slovak, /o/ (for ъ) or /e/ (for ь) in Russian, /a/ in Serbian, etc.

2.2 PIE syllabic resonants and their Slavic reflexes

Proto-Indo-European had four sonorant consonants—**m*, **n*, **l*, **r*—that could function as syllable nuclei in zero-grade ablaut environments, written **m̥*, **n̥*, **l̥*, **r̥*. Each IE daughter branch resolved these syllabic resonants differently, inserting a distinct epenthetic vowel: Sanskrit preserved the syllabic resonant (*r̥*), Greek inserted /a/, Latin /o/, Germanic /u/, Lithuanian /i/, and Slavic wrote jer + liquid (ь*r*, ъ*l*, etc.). The standard reconstruction posits a Proto-Balto-Slavic intermediate stage **iR*/**uR* (Mayer 1991), with ∼73% showing **i* (reflected as front jer ь).

The question at issue is whether this intermediate PBS vowel was a "real" full short vowel that later reduced to a jer (the standard view), or whether Slavic preserved the syllabic resonant more or less directly from PIE, with the jer serving as a scribal device to represent the syllabic function of the consonant (the alternative we develop here).

2.3 Previous treatments of liquid-context jers

Scheer (2003, 2004) provides the most detailed structural analysis of Czech syllabic consonants within a Government Phonology (CVCV) framework. He distinguishes **syllabic consonants** (from Common Slavic ***$t^{\text{ь}}rt$, where the jer precedes the liquid) and **trapped consonants** (from CS ***$tr^{\text{ъ}}t$, where the jer follows). In his framework, the jer is lost and the liquid spreads into the vacated nuclear position. Scheer does not frame the vlk/sen difference as a problem for Havlík's law; in his framework, both outcomes are natural structural consequences of sonorant spreading.

Shevelov (1964) argued that Common Slavic jers were not a special "reduced" category but simply short vowels—the short members of the vowel system. This opens the theoretical space for liquid-context jers to have had a different phonetic realization (minimal or zero vocalic content) from non-liquid jers, even if they were written identically.

Shafirovich (Purdue dissertation) investigated "the possibility of existence of syllabic liquids in Common Slavic" by examining variation between ъ*r*/*t* and *r*ъ*t* spellings in manuscripts, sometimes appearing in the same word without semantic distinction. This is the closest prior work to our argument, but it did not assemble the quantitative case we present here.

Bethin (1998) treats the liquid+jer interaction through the lens of prosodic structure, arguing that syllable reorganization preceded the fall of the jers and that liquid-adjacent jers occupied a distinct prosodic position. In a related vein, Yoo (n.d.), building on Bethin's (1992, 1993) autosegmental analysis of Late Common Slavic syllable structure, argues that the "tense jers" before jod in extreme eastern East Slavic dialects were not genuine jer vowels at all but reflect a different syllabification of the **ŭ*/**ĭ*+jod sequence—a conclusion structurally parallel to ours for a different phonological environment.

Notably, **none of these works explicitly identifies the 100% vs. 0% vocalization asymmetry** documented in the present study, nor draws the inference that liquid-context jers may have been non-vocalic.

3 The Czech data

Czech is the ideal test language for this investigation because it preserves both syllabic /r/ and /l/ as productive phonological categories, making the liquid-adjacent jer outcomes maximally transparent.

3.1 Category A: Strong jers not adjacent to liquids

Table 1 presents all identified cases where a strong-position jer in a non-liquid environment is attested in Proto-Slavic and has a Czech reflex.

Table 1: Strong-position jers in non-liquid environments: Czech outcomes.

#	Proto-Slavic	Gloss	Czech	→ /e/?
1	**sъnъ*	sleep	*sen*	yes
2	**dьnь*	day	*den*	yes
3	**pьsъ*	dog	*pes*	yes
4	**mъxъ*	moss	*mech*	yes
5	**lьnъ*	flax	*len*	yes
6	**vъšь*	louse	*veš*	yes
7	**otьcь*	father	*otec*	yes
8	**konьcь*	end	*konec*	yes
9	**kupьcь*	merchant	*kupec*	yes
10	**šьvъ*	seam	*šev*	yes
11	**lьstь*	cunning	*lest*	yes
12	**dъždžь*	rain	*déšť*	yes
13	**ogňь*	fire	*oheň*	yes
14	**dъska*	board	*deska*	yes
15	**bъzъ*	elder tree	*bez*	yes
16	**pьstrъ*	variegated	*pestrý*	yes
17	**stьblo*	stalk	*stéblo*	yes
18	**dьnьsь*	today	*dnes*	yes
Total strong jers vocalized to /e/:				**18/18**

The result is unambiguous: every strong-position jer in a non-liquid context vocalizes to Czech /e/. The vocalization rate is **100%** (n = 18). Weak-position jers in the same words (**sъ<u>nъ</u>*, **dь<u>nь</u>*) delete as predicted, confirming the standard strong/weak alternation.

Methodological note: The n = 18 figure represents a near-exhaustive survey of Proto-Slavic strong-position jers in non-liquid, non-cluster environments that have transparent Czech reflexes. The small n is a property of the phenomenon, not a sampling limitation: Proto-Slavic had relatively few monosyllabic jer-words, and many have been lost or obscured by later phonological developments. The relevant statistical point is not the sample size but the **categorical nature** of the split—18/18 vs. 0/27—which no reasonable alternative model predicts.

3.2 Category B: Strong jers adjacent to liquids

Table 2 presents strong-position jers adjacent to /r/ (from PIE syllabic _*r̥_), and Table 3 those adjacent to /l/ (from PIE syllabic _*l̥_).

Table 2: Strong-position jers adjacent to /r/: Czech outcomes.

#	Proto-Slavic	Gloss	PIE	Czech	→ /e/?	Outcome
1	_*sъmьrtь_	death	_*mr̥tis_	_smrt_	no	syll. _r_
2	_*mьrtvъ_	dead	_*mr̥twós_	_mrtvý_	no	syll. _r_
3	_*vьrxъ_	top	_*wr̥s-_	_vrch_	no	syll. _r_
4	_*tъrgъ_	market	—	_trh_	no	syll. _r_
5	_*kъrkъ_	neck	_*(s)ker-_	_krk_	no	syll. _r_
6	_*pьrstъ_	finger	_*pr̥sth_$_2$	_prst_	no	syll. _r_
7	_*tьrnъ_	thorn	_*tr̥nós_	_trn_	no	syll. _r_
8	_*pьrvъ_	first	_*pr̥h_$_2$_wós_	_prvý_	no	syll. _r_
9	_*gъrdlo_	throat	_*g_^{w}_r̥h_$_3$_d_^{h}_lom_	_hrdlo_	no	syll. _r_
10	_*zьrno_	grain	_*ǵr̥h_$_2$_nóm_	_zrno_	no	syll. _r_
11	_*sьrdьce_	heart	_*ḱr̥d-_	_srdce_	no	syll. _r_
12	_*čьrvь_	worm	_*k_^{w}_r̥mis_	_červ_	SEC.	OldCz. _črv_
13	_*čьrnъ_	black	_*kr̥snós_	_černý_	SEC.	OldCz. _črný_
14	_*čьrvenъ_	red	← _*čьrvь_	_červený_	SEC.	OldCz. _črvený_
15	_*čьrstvъ_	stale	—	_čerstvý_	SEC.	OldCz. _črstvý_
16	_*bьrvьno_	beam	_*b_^{h}_r̥w-_	_břevno_	SEC.	epenthesis
17	_*čьrta_	line	_*(s)kr̥t-_	_črt_ / _čára_	mixed	_črt_ = syll. _r_

Results for Category B:

- Syllabic consonant preserved (no vocalization at all): **15 words**
- Hard syllabic _l_ → /u/ (not /e/): **4 words**
- Secondary epenthesis (13th–15th c., from Old Czech syllabic consonant): **7 words**
- Direct vocalization to /e/ by Havlík's law: **0 words**

The vocalization rate for liquid-adjacent strong jers is **0%** ($n = 27$).

Table 3: Strong-position jers adjacent to /l/: Czech outcomes.

#	Proto-Slavic	Gloss	PIE	Czech	→ /e/?	Outcome
18	_*vьlkъ_	wolf	_*wl̥k^{w}os_	_vlk_	no	syll. _l_
19	_*pьlnъ_	full	_*pl̥h$_1$nos_	_plný_	no	syll. _l_
20	_*vьlna_	wool	_*h$_2$wl̥h$_1$neh$_2$_	_vlna_	no	syll. _l_
21	_*slьza_	tear	—	_slza_	no	syll. _l_
22	_*žьltъ_	yellow	_*ǵ^{h}l̥h$_3$tos_	_žlutý_	no → /u/	hard l → lu
23	_*dьlgъ_	long	_*dl̥h$_1$g^{h}os_	_dlouhý_	no → /u/	hard l → lu → ou
24	_*dъlgъ_	debt	_*d^{h}elg^h-_	_dluh_	no → /u/	hard l → lu
25	_*tъlstъ_	thick	—	_tlustý_	no → /u/	hard l → lu
26	_*blъxa_	flea	—	_blecha_	SEC.	OldCz. _blcha_
27	_*plъtь_	flesh	_*pleth$_2$-_	_pleť_	SEC.	OldCz. _plt_

Table 4: Havlík's law outcomes by environment (Czech, strong-position jers only).

	Vocalized to /e/	Did not vocalize to /e/
Non-liquid environment	18	0
Liquid-adjacent environment	0	27

3.3 The contingency table

Fisher's exact test on the 2×2 table yields $p < 10^{-12}$. The asymmetry is not a statistical tendency; it is categorical.

4 The secondary epenthesis argument

Seven Category B words show /e/ in Modern Czech: _červ_ 'worm', _černý_ 'black', _červený_ 'red', _čerstvý_ 'stale', _břevno_ 'beam', _blecha_ 'flea', and _pleť_ 'complexion'. A reviewer might object that these represent Havlík's law vocalization and thus constitute counterexamples. Three lines of evidence demonstrate that they do not.

4.1 Old Czech attestation

Historical texts from the 13th–14th centuries attest Old Czech forms **without** /e/: _črv_ (not _červ_), _črný_ (not _černý_), _črvený_ (not _červený_), _črstvý_ (not _čerstvý_), _blcha_ (not _blecha_), _plt_ (not _pleť_), _stblo_ (not _stéblo_). The syllabic consonant stage is directly documented in the written record. The /e/ appeared later.

4.2 Chronological separation

Havlík's law operated during the 10th–12th centuries—the period of the "fall of the jers" across Slavic. The secondary epenthesis that produced *črv* → *červ* and *blcha* → *blecha* occurred in the **13th–15th centuries**, well after the jer fall was complete. These are two distinct sound changes separated by centuries. The /e/ in *černý* was never a jer; it is a later innovation to resolve consonant clusters that had become too complex.

4.3 The /u/ test

When hard syllabic /l/ resolved in Czech, the epenthetic vowel was /u/, not /e/:

**žьltъ*	→ *žlutý*	(not **žletý*)
**dьlgъ*	→ *dlouhý*	(via *dlúhý*, not **dlehý*)
**dъlgъ*	→ *dluh*	(not **dleh*)
**tъlstъ*	→ *tlustý*	(not **tlestý*)

If Havlík's law had vocalized these strong jers, the output should have been /e/—as it is in 100% of Category A cases. The fact that /u/ appears instead proves that a different process was at work: resolution of a syllabic consonant, not vocalization of a jer. The vowel quality tracks the phonetic character of the liquid (back /u/ for velarized /l/), not the quality predicted by jer vocalization.

5 Cross-Slavic evidence

The asymmetry documented in Czech is not a Czech-internal peculiarity. Other Slavic branches show parallel patterns, each in their own way.

5.1 Serbian

Serbian vocalizes non-liquid strong jers to /a/ consistently: **sъnъ* → *san*, **dьnь* → *dan*, **pьsъ* → *pas*. For liquid-adjacent jers, the picture is different:

- **Syllabic /r/ is preserved** across a broad inventory: *smrt* 'death', *prst* 'finger', *crn* 'black', *trn* 'thorn', *trg* 'market', *vrh* 'top', *krv* 'blood', *srp* 'sickle', *grb* 'hump', *brz* 'fast', *grlo* 'throat', *zrno* 'grain', *srce* 'heart', *drvo* 'wood', *tvrd* 'hard', *crv* 'worm', *prvi* 'first'—at least 20 common words, all from PIE syllabic **r̥* or Proto-Slavic ь*r* positions.

- **Syllabic /l/ resolved to /u/**: *vuk* 'wolf' (< *vlk*), *vuna* 'wool' (< *vlna*), *pun* 'full' (< *pln*), *žut* 'yellow' (< *lt*).

The vowel inserted for *l̥ positions is /u/, not /a/. If Havlík's law had vocalized these jers, we would expect /a/—the same reflex as all Category A jers. The /u/ points to a different process: resolution of a syllabic consonant.

A further piece of evidence comes from the 19th-century Serbian spelling reform. When Vuk Karadžić removed jers from the orthography, words like трьнъ became *trn* with **no change in pronunciation**. The /r/ did not "become" syllabic at that point—it already was. The jer had been an orthographic fossil for centuries.

The **asymmetry between /r/ and /l/** is itself informative. Under the standard model, both positions contained "the same jer": why would they behave differently? Under our model, both were syllabic consonants; syllabic /r/ was more typologically stable than syllabic /l/ and survived longer, while /l/ independently resolved to /u/ (completed by ca. 1400, based on manuscript evidence).

5.2 Russian: The polnoglasie connection

Russian presents an initially challenging case for our argument. East Slavic vocalized **all** jers in liquid contexts to full vowels: **vьlkъ* → волк [volk], **sъmьrtь* → смерть [$\mathrm{sm^jert^j}$], **pьrstъ* → перст [$\mathrm{p^jerst}$]. This appears to support the standard model. However, closer examination of the vowel quality reveals that the Russian data *support* rather than contradict our proposal.

5.2.1 The vowel quality anomaly

Under Havlík's law, front jer ь vocalizes to /e/ and back jer ъ vocalizes to /o/. In non-liquid contexts, Russian follows this pattern consistently: **dьnь* → день /e/, **sъnъ* → сон /o/. In liquid contexts, the prediction holds for ь before /r/: **vьrxъ* → верх /e/, **pьrstъ* → перст /e/, **sъmьrtь* → смерть /e/, **zьrno* → зерно /e/, **vьrba* → верба /e/.

But for ь before /l/, the prediction **fails**:

Proto-Slavic	Jer type	Havlík predicts	Russian	Actual vowel	Match?
**vьlkъ*	ь (front)	/e/ (**велк*)	волк	/o/	**no**
**pьlnъ*	ь (front)	/e/ (**пельный*)	полный	/o/	**no**
**dьlgъ*	ь (front)	/e/ (**делг*)	долг	/o/	**no**
**žьltъ*	ь (front)	/e/	жёлтый	/o/ (via *ё*)	**no**

In every case, front jer ь before /l/ yields Russian /o/ instead of the expected /e/. The vowel quality is determined not by the jer color but by the **liquid consonant**: velarized (dark) /l/ conditions backing and rounding of the adjacent vowel, pulling it from the /e/ region to /o/.

5.2.2 The polnoglasie parallel

This vowel-quality mismatch is not an isolated anomaly. It follows a pattern already well known in East Slavic historical phonology: **polnoglasie** (pleophony). East Slavic resolved Proto-Slavic CRC clusters by inserting (copying) a vowel after the liquid:

*gordъ (TORT)	→ город (ToroT)	vowel copy after liquid
*golva (TolT)	→ голова (ToloT)	vowel copy after liquid
*bergъ (TerT)	→ берег (TereT)	vowel copy after liquid
*melko (TelT)	→ молоко (ToloT)	backing before /l/

The last example is critical: _*TelT_ → ToloT. The original /e/ in _*melko_ was backed to /o/ before /l/—the **exact same process** that turned the expected /e/ from ь into /o/ in волк, полный, долг.

The structural parallel is systematic:

Pattern	Input	East Slavic output	Process
Classic polnoglasie	T**or**T	T**oro**T	vowel copy
Liquid-jer (ь+*r*)	T**ь**rT	T**e**rT	/e/ from ь
Liquid-jer (ь+*l*)	T**ь**lT	T**o**lT	/o/ — matches polnoglasie

5.2.3 The geographic isogloss

The geographic split between South Slavic and East Slavic is **identical** for both TORT and TьRT words:

	South Slavic	East Slavic
*gordъ (TORT)	градъ (metathesis)	город (polnoglasie)
*golva (TolT)	глава (metathesis)	голова (polnoglasie)
*vьlkъ (TьlT)	влькъ > Serb. *vuk*	волк (/o/ epenthesis)
*sъmьrtь (TьrT)	Serb. *smrt* (syll. *r*)	смерть (/e/ epenthesis)
*pьrstъ (TьrT)	Serb. *prst* (syll. *r*)	перст (/e/ epenthesis)

In every case, South Slavic preserves or reconfigures the consonant cluster, while East Slavic breaks it up with a full vowel. This is the same phonotactic constraint operating on two successive waves of CRC clusters: the TORT sequences (8th–9th c.) and the TьRT sequences exposed as CRC clusters after the jer fall (10th–12th c.).

5.2.4 Reinterpretation

We propose that what is traditionally described as "vocalization of liquid-context jers" in East Slavic is better analyzed as **polnoglasie applied to syllabic consonants**.

East Slavic had a productive phonotactic constraint against CRC sequences; when the jer fall exposed syllabic consonants as CRC clusters, the same epenthetic mechanism that had resolved TORT sequences was applied to T*l̥*T and T*r̥*T sequences. The vowel quality was determined by the liquid (backing before /l/, preservation before /r/), not by a jer that had already ceased to exist.

Under this analysis, Russian does not constitute evidence *for* the standard model. The East Slavic data are compatible with our proposal: the jers in liquid contexts were non-vocalic, and their apparent "vocalization" was in fact independent epenthesis—the same process, by a different name, that every other IE branch also applied to PIE syllabic resonants.

Further support comes from the Old Novgorod dialect (Zaliznyak 1995), which shows TъRъT / TьRьT patterns—a **copy** of the jer inserted after the liquid, structurally identical to how polnoglasie copies the full vowel. In Novgorod, the two jers apparently *neutralized* in this position, and the resulting vowel quality was conditioned by the consonantal environment rather than the original jer color.

5.3 Polish

Polish resolved liquid-adjacent jers to various vowels: **vьlkъ* → *wilk* [vilk] (/i/), **sъmьrtь* → *śmierć* (/a/ in some forms), **pьlnъ* → *pełny* (/e/). The **diversity of outcomes** (/i/, /e/, /a/) contrasts with the uniform /e/ for non-liquid jers and suggests that the process governing liquid-context resolution was distinct from standard jer vocalization.

5.4 The cross-Slavic summary

Table 5: Cross-Slavic treatment of liquid-adjacent jers vs. non-liquid jers.

Branch	Non-liquid jer →	Liquid jer →	Same process?
Czech/Slovak	/e/	syllabic C	**no**
Serbian	/a/	syll. *r* (kept) / /u/ (*l*)	**no**
Russian	/o/ or /e/	/o/ or /e/ (polnoglasie-like)	**no** (see §5.2)
Polish	/e/	/i/ or /e/ or /a/ (variable)	**no**
Bulgarian	/e/ or /o/	/ъ/ [ɤ] + liquid	**unclear**

In four of the five major branches (Czech, Serbian, Polish, Russian), liquid-adjacent jers **behave differently** from non-liquid jers. For Czech, Serbian, and Polish the distinction is categorical; for Russian, the vowel quality in liquid-adjacent positions follows polnoglasie patterns rather than standard jer vocalization (§5.2). Bulgarian remains unclear but shows its own distinctive treatment (/ъ/ [ɤ] + liquid, distinct from both the /e/ and /o/ of non-liquid jer vocalization).

6 Engagement with the standard account

6.1 The implicit exception

The standard account of Havlík's law does not contain an explicit exception clause for liquid contexts. Instead, textbooks present jer vocalization and syllabic consonant formation as complementary outcomes of the same process, without flagging the logical tension. A representative formulation might read: "Strong jers vocalized; in liquid environments, the liquid became syllabic." But this formulation masks a genuine asymmetry: *why* does the liquid "become syllabic" instead of the jer vocalizing, as it does in every non-liquid context?

The standard answer, insofar as one is given, relies on rule ordering: the jer weakens, the liquid—being the most sonorous adjacent segment—fills the vacated nuclear position. But this is precisely the kind of post hoc explanation that our alternative renders unnecessary.

6.2 Scheer's structural analysis

Scheer's (2003, 2004) CVCV analysis of Czech syllabic consonants is compatible with both the standard and alternative models. In his framework, an empty nucleus (left by the lost jer) is filled by the adjacent liquid through spreading. The synchronic structure is identical whether the nuclear position was once occupied by a real jer or was always empty (with the liquid always associated with the nucleus). The difference between the two models is diachronic—*was there ever a vowel in that position?*—and Scheer's synchronic framework does not adjudicate this question.

What Scheer's analysis does show, importantly, is that syllabic consonants arising from **tъrt* and trapped consonants arising from **trъt* are **structurally distinct** categories, even though both are written with jer in OCS. This supports the broader point that OCS jer notation was not a uniform phonological marker.

6.3 The ъ/ь distinction

One argument for the standard model is that the front/back jer distinction in liquid contexts (ь*r* vs. ъ*r*) implies a vocalic element with quality. A purely syllabic consonant has no front/back dimension.

A systematic survey of 77 Proto-Slavic words with jer adjacent to a liquid (*r* or *l*) substantially weakens this objection. The jer color in these positions turns out to be **etymologically determined**, with near-categorical accuracy.

6.3.1 The front-jer pattern

Among words securely traced to PIE syllabic resonants (*$\mathring{l}$, *$\mathring{r}$), the front jer ь is **universal**: *vьlkъ* 'wolf', *pьlnъ* 'full', *dьlgъ* 'long', *žьltъ* 'yellow', *vьlna* 'wool' (all from *$\mathring{l}$); *mьrtvъ* 'dead', *vьrxъ* 'top', *pьrstъ* 'finger', *zьrno* 'grain', *čьrnъ* 'black', *tьrnъ* 'thorn', *dьržati* 'to hold', *vьrtěti* 'to turn', *žьrti* 'to sacrifice', *četvьrtъ* 'fourth' (all from *$\mathring{r}$). An expanded corpus of 35 such words finds that every one traces to a PIE syllabic resonant (Derksen 2008), with only two inherited exceptions (*vьrba* 'willow', *bьrdo* 'hill'—both with PBS *i* vocalism from non-syllabic roots).

6.3.2 The back-jer pattern

Words with ъ + liquid come from different etymological sources. A corpus of 37 such words reveals four categories, none of which is "PIE syllabic resonant":

- **Loanwords:** *dъlgъ* 'debt' from Gothic *dulgs*; *xъlmъ* 'hill' from PGmc *hulmaz*; *pъlkъ* 'regiment' from PGmc *fulką*; *xъrvatъ* 'Croat' from Iranian.
- **Laryngeal sequences** (*CrH*, not syllabic *$\mathring{r}$): *kъrmiti* 'to feed', *gъrdъ* 'proud', *gъrdlo* 'throat', *gъrnъ* 'furnace'.
- **Full-grade or o-grade roots:** *kъrkъ* 'neck', *plъtь* 'flesh', *zъlъ* 'evil'.
- **Substrate or unknown:** *tъrgъ* 'marketplace', *tъlstъ* 'thick', *blъxa* 'flea'.

Not a single confirmed back-jer + liquid word traces to a PIE syllabic resonant. The distribution is summarized in Table 6.

Table 6: Jer color and etymological source in liquid contexts ($n = 74$).

	PIE *$\mathring{r}$/*$\mathring{l}$	Other source	Total
ь + liquid	**35** (94.6%)	2 (5.4%)	37
ъ + liquid	0 (0%)	**37** (100%)	37

Three loanwords adapted to the ь+liquid pattern (*lьvъ* 'lion', *krьstъ* 'cross', *cьrky* 'church') are excluded; their adaptation to the productive front-jer pattern confirms rather than undermines the diagnostic.

6.3.3 The palatalization override

A revealing contrast confirms the mechanism. Both *čьrvь* 'worm' (from PIE *$k^w\mathring{r}mis$) and *gъrdlo* 'throat' (from PIE *$g^w\mathring{r}h_3d^hlom$) derive from roots with a PIE labiovelar + syllabic resonant, yet the jer colors are **opposite**. In *$k^w\mathring{r}mis$, the labiovelar underwent Slavic first palatalization (*k^w > *č) before the following front vowel *i*, overriding its rounding effect and yielding front jer ь. In *$g^w$$\mathring{r}$$h_3$$d^h$lom*, no palatalization occurred—the labiovelar simply delabialized (*g^w > *g*), and its rounding persisted as back jer ъ. Lithuanian *gurklỹs* 'throat' with *u* (not *i*) confirms the back quality. This is predicted by Kuryłowicz's (1956) rule: after palatalized velars, only PBS **iR* appears.

6.3.4 The diagnostic and its applications

The minimal pair **dьlgъ* 'long' (from PIE *$d\mathring{l}h_1g^h$os*, front jer) vs. **dъlgъ* 'debt' (from Gothic, back jer) epitomizes the pattern: same consonant skeleton, opposite jer color, opposite etymological origin. The jer color in liquid contexts functions as a **diagnostic marker**: back jer ъ signals a non-syllabic-resonant source (loanword, substrate, laryngeal, or full-grade root), while front jer ь signals an inherited PIE syllabic resonant. This diagnostic has predictive power for words of uncertain etymology: **tъrgъ* 'marketplace' (back jer) is predicted to be non-inherited, consistent with its probable paleo-Balkan substrate origin (cf. Illyrian *Tergeste*); **tъlstъ* 'thick' and **blъxa* 'flea' (both back jer) are predicted not to derive from syllabic resonants, consistent with their lack of clear PIE cognates. More broadly, any word with ъ before a liquid that is currently classified as "inherited from PIE syllabic resonant" warrants re-examination.

The overwhelmingly front-jer pattern among genuine syllabic resonant words corresponds to the PBS **iR*/**uR* split, where ∼73% of cases show **i* (Mayer 1991). The front jer is the **default, automatic** reflex; the back jer signals a different etymological source (loanword, laryngeal, or the minority **uR* class). The jer thus encodes inherited PBS-level information, not independent Slavic vocalic quality—exactly what one would expect of a notational system that preserves traditional spelling distinctions from an earlier stage.

6.4 The Balto-Slavic agreement

Baltic and Slavic agree on front (**i*) vs. back (**u*) quality in the same liquid-context words (∼73% **i*; Mayer 1991), suggesting a shared PBS innovation rather than independent development. This is genuine evidence for a PBS stage with *some* vocalic quality in these positions. We do not dismiss this evidence.

However, the agreement is equally compatible with a PBS stage where the resonant was quasi-syllabic with incipient vocalic coloring—not a full vowel. Baltic developed this coloring into a full vowel **i* (Lithuanian *vilkas*, *vilna*, *pilnas*); Slavic preserved the syllabic consonant with the inherited coloring encoded in the ъ/ь spelling (see §6.3). The agreement on front/back quality requires a shared starting point, but that starting point need not have been a full short vowel.

Furthermore, the Balto-Slavic agreement on quality distribution is itself informative for our argument. Mayer's (1991) comprehensive study of 215 PBS lexical items shows that the front (*i) reflex accounts for ~73% of cases, back (*u) for ~17%, and both for ~10%. If the PBS vowel had been a robust, phonologically active short vowel, we would expect the conditioning of its front/back quality to be well understood. In fact, over a century of scholarship has failed to establish the conditioning definitively (proposals by Vaillant, Kurylowicz, Endzelīns, and Stang all capture only a subset of cases). This opacity is more consistent with a quasi-syllabic resonant whose incipient coloring was phonetically variable than with a firmly established short vowel.

7 The proposal

7.1 Two types of jer

We propose that the OCS graphemes ъ and ь served at least three distinct functions, two of which are phonologically genuine and one of which is notational:

Table 7: Three functions of the jer graphemes.

Function	Example	PIE source	Real vowel?
Reduced vowel (< PIE *u)	сънъ 'sleep'	**supnos*	**yes**
Reduced vowel (< PIE *i)	дьнь 'day'	**dyew-*	**yes**
Syllabic C notation	вльkъ 'wolf'	*$w\underset{\circ}{l}k^w os$*	**no**
Loanword epenthesis	стькло 'glass'	Gmc **stikls*	**no**

Havlík's law applies to the first two categories: genuine reduced vowels that alternate between vocalization (strong position) and deletion (weak position). It does not apply to the third category, because there is no vowel to vocalize. The 100% vs. 0% split in the Czech data (Tables 1–3) reflects this categorical distinction.

7.2 A reformulated Havlík's law

Under the standard formulation, Havlík's law requires an implicit exception: "Strong jers vocalize—*except in liquid environments, where the liquid becomes syllabic instead.*" Under our reformulation, no exception is needed:

> **Havlík's law (reformulated):** In Proto-Slavic words containing genuine reduced vowels (ъ from *u, ь from *i), weak-position jers are deleted and strong-position jers vocalize to full vowels. Orthographic jers representing syllabic consonants (from PIE *$\underset{\circ}{l}$, *$\underset{\circ}{r}$) are outside the domain of the law.

This reformulation makes Havlík's law **more regular**, not less. The apparent exception for liquid contexts vanishes because the jers in those positions were never part of the law's input.

7.3 Implications for Proto-Balto-Slavic

If liquid-context jers were not real vowels, then the standard PBS reconstruction requires revision. Instead of:

$$\text{PIE } {*}w\mathring{l}k^{w}os \rightarrow \text{PBS } {*}wilkas \rightarrow \text{PS } {*}v\text{ъ}lk\text{ъ} \rightarrow \text{Czech } vlk$$

we have:

$$\text{PIE } {*}w\mathring{l}k^{w}os \rightarrow \text{PS } {*}v\mathring{l}k\text{ъ} \text{ (syllabic } l \text{ preserved)} \rightarrow \text{Czech } vlk$$
$$\text{PIE } {*}w\mathring{l}k^{w}os \rightarrow \text{PB } {*}wilkas \text{ (independent } i\text{-epenthesis)} \rightarrow \text{Lithuanian } vilkas$$

One sound change per branch, rather than two changes in Slavic (epenthesis, then reduction back to syllabic consonant). This is more parsimonious and consistent with the typological profile of Slavic as a phonologically conservative branch of IE (seven cases, dual number, rich verbal aspect, preservation of initial **w-*).

We acknowledge that the PBS question extends beyond the scope of this paper and requires engagement with the broader Balto-Slavic reconstruction literature. The Havlík's law data presented here is necessary but not sufficient for revising the PBS reconstruction; it is, however, a concrete empirical argument that the standard model must address.

8 Slavic-internal divergence as continued lability

A further observation connects this paper to the broader study of PIE phonological instability (Jevremović 2026). Modern Slavic branches independently resolve syllabic consonants in different ways, producing a second round of divergence within a single language family:

Table 8: Modern Slavic reflexes of PIE $*w\mathring{l}k^{w}os$ 'wolf'.

Language	Reflex	Epenthetic vowel
Czech	*vlk*	∅ (syllabic *l*)
Serbian	*vuk*	/u/
Russian	волк	/o/
Polish	*wilk*	/i/
Bulgarian	вълк	/ъ/

Five branches produce five different vowels at the same position. Under the standard model, this is "vocalization of an inherited reduced vowel"—but then the question arises why a single reduced vowel produced five different outcomes. Under our model, this is independent epenthesis in each branch—the same process that operated across the full

IE family at these positions (Sanskrit *ṛ*, Greek /a/, Latin /o/, Germanic /u/, Lithuanian /i/), now recurring at the Slavic level. The instability is inherent to the phonological position, not to the particular vowel that happened to occupy it.

9 Open questions

We flag several questions that require further research:

1. **Non-PIE-syllabic-resonant liquid jers.** Are there Proto-Slavic words where jer + liquid arises from a source other than PIE **l̥/*r̥*? If such words exist, their Czech outcomes would be diagnostic: if they show /e/ vocalization, this would confirm that the Category A/B split tracks PIE origin, not just phonological context.

2. **Old Russian manuscripts.** Is there evidence for a syllabic consonant stage in Old Russian *before* full vocalization? If East Slavic manuscripts show variation between jer + liquid and liquid-only spellings (as OCS manuscripts do), this would support a syllabic consonant intermediate stage.

3. **Slovak phonemic length.** Slovak preserves syllabic /r/ and /l/ with phonemic *length* (*vĺča* 'wolfling', *kĺb* 'joint'). The existence of length contrasts on syllabic consonants raises questions about the vocalic content of these segments.

4. **Formal statistical modeling.** While the 2×2 contingency table (Table 4) is sufficient for demonstrating the categorical nature of the split, a mixed-effects logistic regression controlling for word frequency, syllable structure, and the specific liquid involved would strengthen the analysis.

10 Conclusion

We have documented a categorical asymmetry in the outcomes of Havlík's law in Czech: strong-position jers vocalize to /e/ in 100% of non-liquid cases ($n = 18$) but in 0% of liquid-adjacent cases ($n = 27$). The seven Modern Czech words with /e/ near liquids are secondary epenthesis, documented in Old Czech as syllabic consonant forms, and separated from Havlík's law by two to three centuries. The resolution of hard syllabic /l/ to /u/ (not /e/) confirms a process distinct from jer vocalization. Cross-Slavic data from Serbian and Polish corroborate the pattern. In Russian, the vowel quality at liquid-jer positions follows polnoglasie patterns (front jer ь → /o/ before /l/, matching **TelT* → ToloT) rather than standard jer vocalization, indicating that East Slavic "vocalization" was independent epenthesis to resolve syllabic consonants, not Havlík's law.

We argue that this asymmetry constitutes evidence that liquid-context jers in OCS were notational conventions for syllabic consonants, not genuine reduced vowels. Havlík's law, reformulated as applying only to real reduced vowels (from PIE **u/*i*), becomes more regular: the "liquid exception" disappears because there was never a vowel to except.

The argument has implications for Proto-Balto-Slavic reconstruction: if Slavic preserved syllabic resonants directly from PIE, the standard PBS **iR/*uR* stage may represent an innovation of Baltic alone, with Slavic and Baltic independently resolving PIE syllabic resonants—Slavic by preserving them, Baltic by epenthesizing **i*. This more parsimonious account aligns with the broader typological conservatism of Slavic.

Our proposal does not invalidate Havlík's law. It refines its domain: the law correctly describes the behavior of genuine reduced vowels, which is exactly the category it was designed to explain. What it does not describe is the behavior of positions that never had a vowel to begin with.

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